ALPACA AI TRADING AGENTS HACKATHON · LABLAB.AI · 28 AUG – 4 SEP 2026

IV DESK

An autonomous options desk that sells overpriced volatility — and stands down the rest of the time.

Paper trading only

Account PA39HSCQE8S3

github.com/alvaarocl/iv-desk

THE IDEA

An insurance company for the stock market

An insurer prices risk correctly and refuses reckless customers — and wins over time. IV Desk does the same thing to index options: it sells "insurance" (premium) on SPY, QQQ and IWM, only when that insurance is statistically overpriced.

- → Defined-risk structures only — iron condors, 1–3 days to expiration
- → Sells only when implied vol sits above a realized-vol forecast (VRP)
- → Sells only when dealer gamma favors range-bound behaviour

What makes this different

Most agents at this hackathon predict price direction. This one doesn't try. It reads the **option surface**, not the chart — and the product isn't the trades. It's the **refusals**, each one written to an append-only journal with the exact number that caused it.

295 documented stand-downs this week. 1 trade. Both are the strategy working.

ARCHITECTURE

The LLM never touches the money path

LAYER	DETERMINISTIC?	WHAT IT DOES
Signal	yes	VRP ratio (IV vs. RV forecast), normalized dealer gamma, regime, skew
Structure & sizing	yes	Strike selection at a target delta behind liquidity gates; contracts capped to the risk budget
Risk Officer	yes	Every gate, no discretion — `evaluate()` returns (ok, reason), no LLM path in
The desk (LLM)	bounded	Four seats debate *whether* to open a trade already approved — can only trim, veto, or do nothing
Execution & exits	yes	4-leg `mleg` order via the Alpaca CLI; 50% TP / 2× stop / expiry close
Journal	yes	Append-only JSONL — every signal, refusal, debate transcript, open and exit

THE LINE THAT MAKES IT STRUCTURALLY SAFE — NOT A PROMPT PROMISE

```
contracts = max(0, min(int(head.contracts), cap)) # agent/debate.py
```

THE SIGNAL

Three gates. Any one stands the desk down.

① Volatility risk premium

ATM implied vol over a Yang-Zhang + EWMA realized-vol forecast, expressed as a **ratio** — holds across any vol regime. Gate: ratio ≥ 1.05.

② Dealer gamma (GEX)

Normalized from chain open interest (T-2 data — a regime signal, never a precision input), with a dead zone near zero. Positive gamma = dealers dampen moves.

③ Regime

ADX / EMA read. A trending tape stands the desk down rather than fading it — short premium into a real trend is how short-vol books die.

Four seats, all open models, all on Featherless

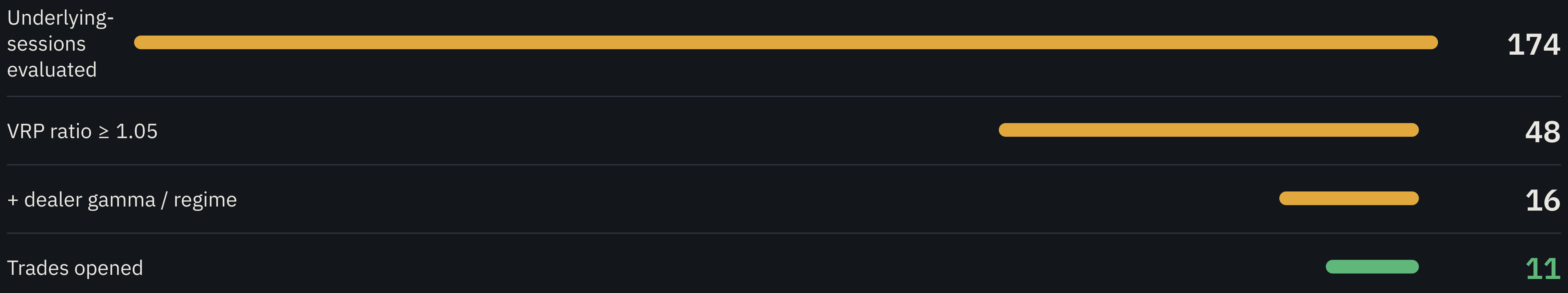
- **Quant ensemble** — up to 3 open models vote independently; needs a strict majority *of the models dispatched*, not of those that answered
- **Bull / Bear** — must anchor claims to real numeric signal fields, named; an argument citing fewer than two is discarded outright
- **Desk Head** — sizes the trade (≤ the Risk Officer's cap) and writes a **falsifiable prediction**: a closing range, on a date

No path where garbage means yes

Outage, timeout, truncated JSON, a split ensemble — every failure mode resolves to `approved=False`, reason recorded. 90-second wall-clock budget for the whole debate, because the tick is 15 minutes. `DESK_DEBATE=off` is the kill switch — the default is `required`: no working LLM layer means no new positions, period.

DISCIPLINE — THE PART WE'RE PROUDEST OF

60 real sessions, 174 opportunities, 11 trades



+$484
HELD-TO-EXPIRY P&L

8 / 11
WINNERS (BACKTEST)

78 → +$401
WITH EVERY GATE DISABLED — EDGE IS REAL, BUT THIN

No honest lever buys more trades without hurting the expectancy. This is a strategy whose **discipline is auditable** — every one of the other 163 sessions is in the journal with the gate that blocked it.

NOT A CLAIM — TWO REAL VETOES FROM THIS WEEK

Standing down, not rationalizing

2 Sep — SPY cleared VRP, gamma vetoed it

Ratio 1.61 (well above the 1.05 floor) — but GEX read −0.106 (dealers short gamma), so the signal layer stood it down before the desk was even due to see it. It ran anyway, in an observation-only **shadow debate**: 2 of 3 Quant models confirmed, Bull and Bear argued genuinely opposite cases, and the Desk Head overruled its own Quant majority.

> *"The GEX signal must be strong enough to overcome the high ATM IV and choppy regime for this trade to win."*

3 Sep — every signal gate cleared, the credit floor didn't

VRP 1.74, GEX +0.58 (long gamma — exactly the regime the strategy sells into), sized to 7 contracts by the Risk Officer — and still rejected before reaching the desk: credit/width came back at 0.10 against the 0.20 floor. A 0DTE condor simply doesn't carry the credit that floor was calibrated on.

RESULTS — THE SCORED WINDOW, MON 31 AUG → THU 3 SEP CLOSE

1 trade, 1 win, +$318.85

ENDING EQUITY

$100,318.85

+$318.85 · +0.32% · four sessions

QQQ iron condor

8 contracts · strikes 713/715P – 720/722C · $352 credit · $1,248 max loss. Opened Thu 3 Sep 11:58 ET after clearing every gate *and* the real (non-shadow) four-seat debate — the mesa's only live approval of the week. Closed inside every strike.

295	360	4	1 / 1
DOCUMENTED STAND-DOWNS	SIGNAL EVALUATIONS THIS WEEK	DEBATES RUN, 1 APPROVAL	FALSIFIABLE PREDICTIONS CORRECT

Desk Head's prediction: *QQQ closes in [705, 730] on 2026-09-03* — correct.

181 tests — and the bug we didn't hide

- Four-seat debate tested end-to-end against injected transport doubles — no network, no API keys
- Black-Scholes IV/greeks backfill (23 tests) — built mid-week when Alpaca turned out not to publish greeks for same-day options
- Strike selection, exit manager, risk gates and the account guard all tested against synthetic data

Issue #26 — the real incident

`_close_legs` flipped a leg's `side` but not its `position_intent` prefix. The one real trade this week hit it live — 13 failed close attempts, 13:28–15:13 ET — before the position expired inside the strikes and OCC settled it automatically.

Root-caused from the live error log, fixed, and pinned with 4 regression tests the same afternoon. Left in the record, not quietly reverted — a desk that documents its own bugs the same way it documents its stand-downs is the point of this project.

ALPACA STACK USAGE

Deep, real use of the platform

- **Trading API through the official CLI** — every trading call in the production loop shells out to `alpaca api METHOD /path`, pinned and version-verified in CI
- **order_class: mleg** — a 4-leg iron condor as one order, signed `limit_price` (negative = credit), zero-price rejected outright
- **Options market data over REST** — chain snapshots with greeks and IV, per-contract open interest, SIP daily bars
- **Idempotent and self-healing** — deterministic `client_order_id` per trade intent; every tick reconciles against Alpaca as the source of truth
- **Paper only** — competition account PA39HSCQE8S3, $100,000, first order Mon 31 Aug 09:30 ET, untouched before that

CLOSE

The journal is in the repo — every signal, every refusal, every trade.

github.com/alvaarocl/iv-desk

alvaarocl.github.io/iv-desk

Account PA39HSCQE8S3